

REPORT

CUSTOMER CHURN ANALYSIS AND PREDICTION

Data Analysis Internship Project

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Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, Jupyter Notebook

Abstract

Customer churn is a significant challenge for telecommunication companies because losing customers directly impacts revenue and growth. This project focuses on analyzing customer behavior and developing a machine learning model to predict customer churn. Various data analysis and visualization techniques were used to identify important factors influencing customer attrition. A Logistic Regression model was developed to predict customer churn with an accuracy of approximately 81.5%.

1. Introduction

Customer churn refers to customers discontinuing their services with a company. Understanding the reasons behind churn helps businesses design effective retention strategies and improve customer satisfaction.

The objective of this project is to:

- Analyze customer behavior patterns.
 - Identify factors affecting churn.
 - Build a predictive machine learning model.
 - Recommend strategies for improving customer retention.
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2. Dataset Description

The project uses the Telco Customer Churn dataset.

Dataset Information

- Total Customers: 7043
- Number of Features: 21
- Target Variable: Churn

Major Features

- Gender

- Senior Citizen
 - Partner
 - Dependents
 - Tenure
 - Internet Service
 - Contract
 - Payment Method
 - Monthly Charges
 - Total Charges
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3. Data Cleaning and Preprocessing

Several preprocessing techniques were performed to prepare the dataset for analysis:

Missing Value Handling

Missing values present in the dataset were identified and appropriately handled.

Duplicate Removal

Duplicate records were checked and removed to maintain data quality.

Data Type Conversion

Necessary data types were converted to suitable numerical formats.

Label Encoding

Categorical variables were transformed into numerical values to make them suitable for machine learning algorithms.

Outcome

A clean and structured dataset was obtained for further analysis and predictive modeling.

4. Exploratory Data Analysis

Various visualizations were created to understand customer behavior.

Customer Churn Distribution

- 73.5% customers were retained.
- 26.5% customers churned.

Contract Type Analysis

Month-to-month customers exhibited the highest churn rate.

Customers having one-year and two-year contracts showed better retention.

Internet Service Analysis

Fiber optic customers showed higher churn compared to DSL users.

Payment Method Analysis

Customers using electronic checks experienced relatively higher churn rates.

Partner and Dependents Analysis

Customers without partners and dependents tended to churn more frequently.

5. Customer Segmentation

Customers were segmented based on:

Tenure Groups

- 0–12 Months
- 13–24 Months
- 25–48 Months
- 49–72 Months

Monthly Charge Groups

- Low
- Medium
- High

Findings

- Short-tenure customers exhibited higher churn.
 - High monthly charge customers were more likely to leave.
 - Long-tenure customers showed stronger retention.
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6. Correlation Analysis

Correlation heatmaps were generated to study relationships among numerical variables.

Observations

- Tenure had a negative relationship with churn.
- MonthlyCharges had a weak positive relationship with churn.
- TotalCharges showed a negative relationship with churn.

These relationships provide valuable insights into customer behavior.

7. Customer Lifetime Value Analysis

Customer Lifetime Value (LTV) was analyzed to understand customer profitability.

Findings

- Most high-value customers remained with the company.
- Only a smaller proportion of valuable customers churned.

This indicates that retaining valuable customers can significantly improve business performance.

8. Machine Learning Model

Logistic Regression

A Logistic Regression classifier was implemented for customer churn prediction.

Steps Followed

1. Train-Test Split
2. Model Training
3. Prediction
4. Model Evaluation

Evaluation Metrics

- Accuracy
 - Precision
 - Recall
 - F1-score
 - Confusion Matrix
 - Classification Report
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9. Model Performance

The confusion matrix produced the following results:

Actual / Predicted No Churn Churn

No Churn	934	102
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Actual / Predicted No Churn Churn

Churn	159	214
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Accuracy Calculation

Accuracy = $(934 + 214) / 1409$

Model Accuracy

81.5%

The model demonstrated satisfactory performance and effectively distinguished between churned and retained customers.

10. Factors Influencing Churn

According to Logistic Regression coefficients, the top factors affecting churn are:

1. Paperless Billing
2. Internet Service
3. Senior Citizen
4. Multiple Lines
5. Charge Group
6. Partner
7. Payment Method
8. Streaming Movies
9. Monthly Charges
10. Total Charges

These variables have the greatest influence on customer churn prediction.

11. Customer Retention Strategies

Based on the analysis, the following recommendations are suggested:

Loyalty Programs

Reward long-term customers to increase retention.

Personalized Offers

Provide targeted discounts and promotional plans.

Improved Customer Support

Enhance service quality and address customer issues promptly.

Long-Term Contracts

Encourage customers to adopt one-year and two-year plans.

High-Risk Customer Monitoring

Identify customers likely to churn and implement preventive measures.

12. Key Findings

- ✓ 73.5% customers were retained while 26.5% churned.
 - ✓ Month-to-month contracts showed the highest churn rate.
 - ✓ Short-tenure customers were more likely to leave.
 - ✓ Customers with higher monthly charges exhibited increased churn.
 - ✓ Fiber optic users experienced relatively higher churn.
 - ✓ Electronic check users had higher churn rates.
 - ✓ Long-term contracts improved customer retention.
 - ✓ The Logistic Regression model achieved approximately 81.5% accuracy.
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13. Conclusion

This project successfully analyzed customer behavior and developed a predictive model for customer churn.

The analysis revealed that contract type, tenure, monthly charges, internet service, and payment methods significantly influence customer attrition. The Logistic Regression model achieved approximately 81.5% accuracy, demonstrating its effectiveness in predicting customer churn.

These findings can help telecommunication companies improve customer retention strategies and reduce revenue loss.

14. Future Scope

Future improvements may include:

- Random Forest Classifier
- Decision Tree Classifier
- XGBoost Algorithm

- Deep Learning Models
 - Hyperparameter Tuning
 - Power BI Dashboard Development
 - Deployment using Streamlit or Flask
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References

1. Telco Customer Churn Dataset
 2. Pandas Documentation
 3. Scikit-Learn Documentation
 4. Seaborn Documentation
 5. Matplotlib Documentation
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Thank You

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